

Exp. No: 5 Determination and plotting of Electric potential difference between any two points in free space medium separated by a distance R.

Program:

```
1 clc;
2 clear;
3 epsilon0 = 8.854e-12;
4 pi = %pi; ...//use-Scilab's-built-in-pi
5 //Input
6 q = input("Enter the q-value: ");
7 ax = input("Enter the value of ax=");
8 ay = input("Enter the value of ay=");
9 bx = input("Enter the value of bx=");
10 by = input("Enter the value of by=");
11 //Distances
12 r1 = sqrt(ax^2 + ay^2);
13 r2 = sqrt(bx^2 + by^2);
14 //Potentials
15 v1 = q / (4 * pi * epsilon0 * r1);
16 v2 = q / (4 * pi * epsilon0 * r2);
17 //Potential difference
18 dv = v1 - v2;
19 //Output
20 disp("The value of v1 =" + string(v1));
21 disp("The value of v2 =" + string(v2));
22 disp("The electric potential difference =" + string(dv)
23 );
24 //Plotting
25 r = linspace(0, 10, 50);
26 y = v1 * exp(-0.1 * r);
27 z = v2 * exp(-0.1 * r);
28 x = dv * exp(-0.1 * r);
29 figure();
30 subplot(2,2,1);
31 plot(r, y);
32 xlabel("Radius-1");
33 ylabel("Voltage-1");
34 title("Potential-V1");
```

Output:

Enter the q value: 2×10^{-6}

Enter the value of ax= 2

Enter the value of ay= 3

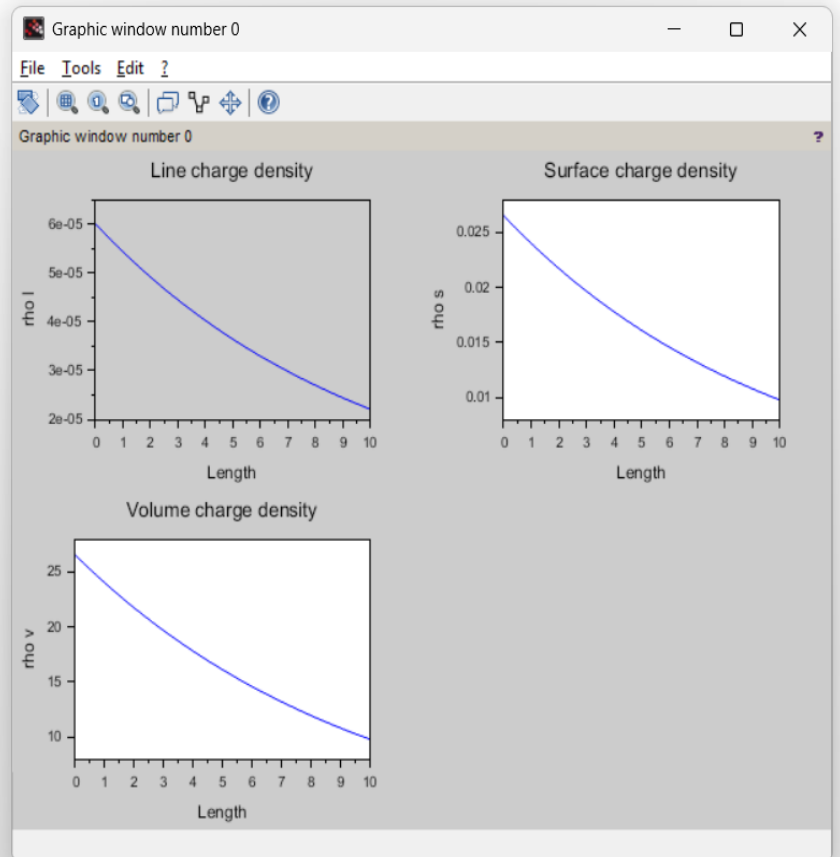
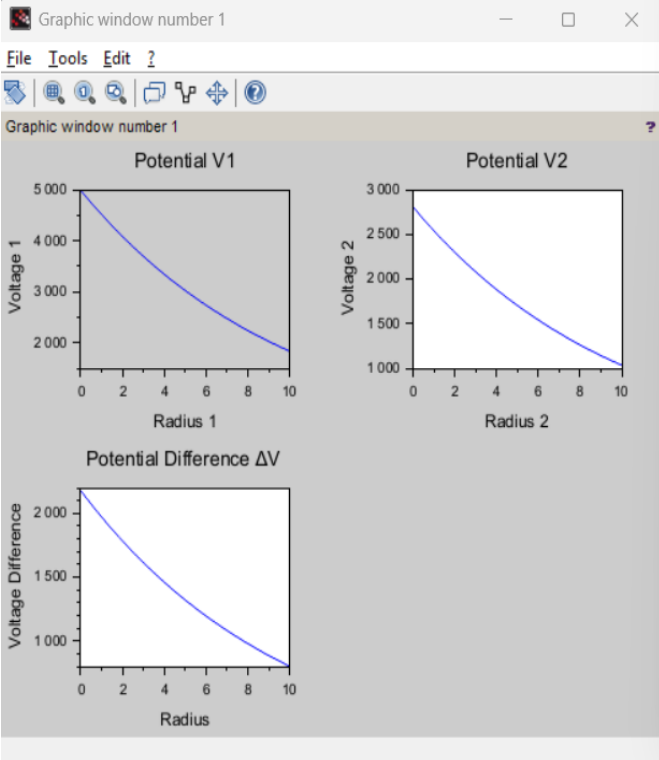
Enter the value of bx= 4

Enter the value of by= 5

"The value of v1 = 4985.5025"

"The value of v2 = 2807.2991"

"The electric potential difference = 2178.2034"



Calculation:

Exp: 5

Enter q value: 2×10^{-6}

Enter value of ax: 2

Enter value of ay: 3

Enter value of bx: 4

Enter value of by: 5

A) From the given values

$$\epsilon_0 = 8.854 \times 10^{-12}$$

$$\pi = 3.14$$

i) Distance

$$r_1 = \sqrt{(ax)^2 + (ay)^2} = \sqrt{2^2 + 3^2} = \sqrt{13} = 3.606$$

$$r_2 = \sqrt{(bx)^2 + (by)^2} = \sqrt{4^2 + 5^2} = \sqrt{41} = 6.403$$

ii) Potentials:

$$V_1 = \frac{q}{4\pi\epsilon_0 r_1} = \frac{2 \times 10^{-6}}{(4 \times 3.14 \times 8.854 \times 10^{-12})(3.606)} = 4985.502 //$$

$$V_2 = \frac{q}{4\pi\epsilon_0 r_2} = \frac{2 \times 10^{-6}}{(4 \times 3.14 \times 8.854 \times 10^{-12})(6.403)} = 2807.299 //$$

iii) Potential difference:

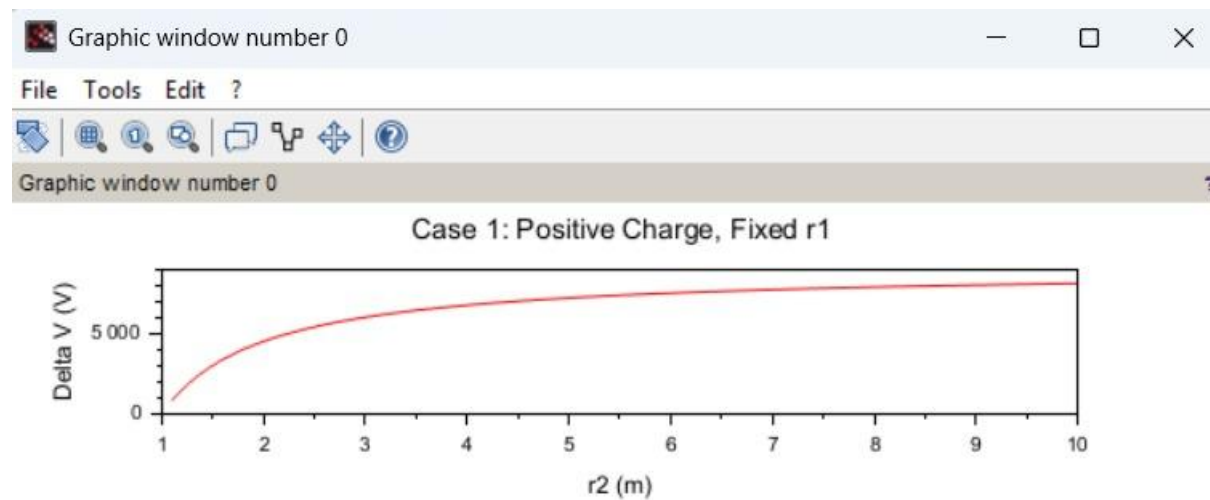
$$\Delta V = V_1 - V_2 = 4985.502 - 2807.299$$

$$= 2178.2034 //$$

Case 1: A positive charge with fixed $r_1=1$ m r_2 varying.

```
1 //Constants
2 epsilon0 = 8.854e-12; //Permittivity of free space
3 k = 1./ (4 * %pi * epsilon0); //Coulomb's constant
4 q = 1e-6; //Charge in Coulombs (1 µC)
5
6 //Case 1: Positive charge, r1=1 m, r2 varies
7 r1_case1 = 1; //meters
8 r2_case1 = linspace(1.1, 10, 100); //meters
9 deltaV_case1 = k * q * (1./ r1_case1 - 1./ r2_case1);
10
11
12
13 //Plotting
14 clf();
15 subplot(3, 1, 1);
16 plot(r2_case1, deltaV_case1, "r");
17 xlabel("r2 (m)");
18 ylabel("Delta V (V)");
19 title("Case 1: Positive Charge, Fixed r1");
20
21
```

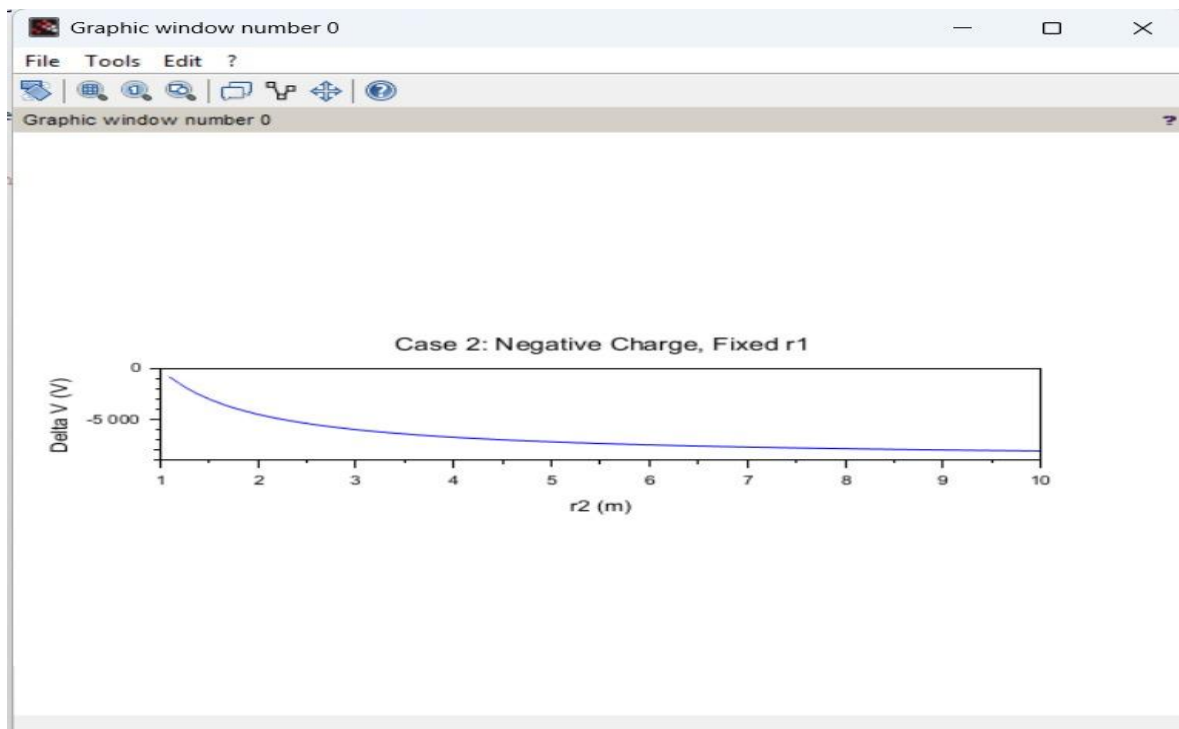
Output:



Case 2: A negative charge with r_1 fixed and r_2 varying.

```
exp 5.sci x exp 5 case 1.sci x exp 5 case II.sci x
1 //Case 2: Negative charge, r1 fixed, r2 varies
2 q_case2 = -1e-6; // -1 μC
3 r1_case2 = 1; // meters
4 r2_case2 = linspace(1.1, 10, 100); // meters
5 deltaV_case2 = k * q_case2 * (1./ r1_case2 - 1./ r2_case2);
6 //Plotting
7
8 clf();
9 subplot(3, 1, 2);
10 plot(r2_case2, deltaV_case2, "b");
11 xlabel("r2 (m)");
12 ylabel("Delta V (V)");
13 title("Case 2: Negative Charge, Fixed r1");
14
```

Output:



Case 3: Both points r1_r1 and r2_r2 vary with a positive charge.

```
exp 5 case #.sci (C:\Users\jasy\OneDrive\Documents\exp 5 case #.sci) - SciNotes
exp 5.sci  exp 5 case 1.sci  exp 5 case #.sci  exp 5 case #.sci
1 // Case 3: Positive charge, both r1 and r2 vary
2 r1_case3 = linspace(0.5, 5, 100); // meters
3 r2_case3 = linspace(5, 10, 100); // meters
4 deltaV_case3 = k * q * (1 ./ r1_case3 - 1 ./ r2_case3);
5 // Plotting
6
7 clf();
8 subplot(3, 1, 3);
9 plot(r1_case3, deltaV_case3, "g");
10 xlabel("r1 (m)");
11 ylabel("Delta V (V)");
12 title("Case 3: Positive Charge, Both r1 and r2 Vary");
13
14
```

Output:

